(Preprint) AAS XX-XXX

WARNING TIME OPTIMISATION OF LOW-THRUST SATELLITE COLLISION AVOIDANCE VIA DIFFERENTIAL ALGEBRA

Frank de Veld*, Roberto Armellin†, Zeno Pavanello‡

The increasing density of space debris necessitates frequent and efficient collision avoidance maneuvers (CAMs) for operational satellites. This work presents a novel methodology that determines the latest possible thrust initiation time of low-thrust CAMs with differential algebra, offering a balance in accuracy and computational efficiency making it suitable for on-board CAM optimisation. In particular, this work dynamically tracks the time of maximum conjunction danger as a function of the control input, and leverages this tracking to find the latest possible time at which a thrust arc can be initiated while safely avoiding a collision between an operational satellite and a non-cooperative object.

INTRODUCTION

Recently, the increasing presence of space debris in Earth-centered orbits has posed significant challenges for satellite operators. As space debris objects orbit the Earth uncontrollably, they can pose a threat of collision to actively operated satellites nearby. Due to the high relative speeds involved, any collision would cause significant damage to satellites. Satellites are therefore forced to perform collision avoidance manoeuvres (CAMs) to stay clear of a collision when a close encounter is predicted to occur.

Many satellites in crowded operational orbits primarily use low-thrust propulsion systems, such as ion engines, Hall Effect thrusters and electrospray propulsion. Although these systems are more efficient than high-thrust propulsion, they reduce manoeuvrability, and manoeuvres such as CAMs can require minutes or even hours of thrust arcs. This limitation makes it important to start a satellite thrust arc adequately in advance, and can make optimisation of these arcs a complex task.

Existing literature primarily focuses on optimising CAM thrust arcs for fuel consumption or Δv, typically resulting in early maneuvers. However, operational constraints often favor delaying CAMs as long as possible to leverage updated tracking data and reduce unnecessary maneuvers. The danger of a satellite-debris conjunction is often assessed in terms of the probability of collision (PoC), a metric based on the propagated position uncertainty distribution. This uncertainty is regularly updated through ongoing tracking of the debris object. More recent tracking data results in less positional uncertainty and therefore a more accurate assessment of conjunction danger and whether a CAM is needed or not.

Therefore, this work focuses on optimising the warning time for the CAM, that is, investigating the latest possible moment in time at which a thrust arc can be initiated to safely avoid a collision

*PhD Student, Department of Mathematics, INRIA/Université Côte d'Azur, 06000, Nice, France.
†Professor, Te Puñaha Ātea Auckland Space Institute, The University of Auckland, 1010, Auckland, New Zealand
‡??

(i.e., decreasing the PoC to a safe threshold). Not only is this kind of optimisation pertinent to collision alerts which are received very late, but it can also be used to investigate whether there is enough time left to wait for the next batch of debris tracking data to arrive and re-assess whether a CAM is necessary. Earlier, this problem has been investigated in our prior work from an optimal control standpoint.[1] While rigorous, it can be complicated to include for example inequality constraints in this methodology

Existing techniques for low-thrust CAM optimisation often rely on numerical integration, which can be computationally intensive, or analytical approximations, which may sacrifice accuracy. For example, Pavanello et al. explored recursive polynomial methods for designing fast low-thrust collision avoidance manoeuvres, emphasising the potential for continuous optimisation of thrust profiles.[2] Mahfouz et al. investigated fuel-optimal formation reconfiguration using unidirectional low-thrust propulsion systems, focusing on minimising fuel consumption through advanced trajectory optimisation techniques, which is a different objective than optimising warning time.[3] Paduraru et al. applied reinforcement learning to optimise low-thrust CAMs, demonstrating improvements in computational efficiency compared to traditional methods, but the computational demands may be prohibitive for on-board implementation.[4]

Despite these advances in CAM optimisation, limitations such as the high computational cost of numerical methods and the potential inaccuracies of analytical approximations persist. Differential Algebra (DA) offers a promising alternative by efficiently propagating state and uncertainty information through Taylor series expansions. This allows for fast and accurate computation of the state and its derivatives, crucial for collision avoidance in dynamic and uncertain orbital environments, and they are flexible regarding the inclusion of operational constraints. This work leverages DA to address these challenges, aiming to enhance both the precision and computational feasibility of low-thrust CAM optimisation, specifically the optimisation of warning time.

Pavanello et al. have previously investigated low-thrust CAM optimisation, enabling efficient computation of the uncertainty distribution of objects involved in a conjunction at any point in time. Their approach integrates seamlessly with advanced higher-order methods, such as second-order cone programming, to address the non-convex nature of CAM optimisation problems.[5] Building on the DA-based uncertainty propagation methods developed by Pavanello et al., this work introduces a novel approach by treating the time of maximum danger as a variable rather than a fixed parameter, as is commonly done. By allowing the time of maximum danger to vary, this methodology provides greater flexibility and accuracy and enables manoeuvres to shift the time of maximum danger further into the future.

In recent years, more attention has been paid to actively removing dangerous debris objects from orbit, but this work is still in its first steps. With more and more active satellites in orbit and enhanced debris tracking capabilities increasing the number of debris objects tracked, the number of CAMs required for safe space operations will only increase in coming years. Industry is working towards automated on-board collision avoidance, for which robust and efficient CAM optimisation algorithms are required. The DA methodology introduced in this work can contribute significantly to this goal.

This work is organised into a number of sections. First, the dynamics of the system and the close approach are presented, with the second part applying the DA techniques to the problem. Results about the performance of the methodology are then presented, as well as some strengths and weaknesses of the approach, followed by a conclusion.

PROBLEM DYNAMICS

We consider a controlled primary satellite described by a state

$$x_p = \begin{bmatrix} r_p \\ v_p \end{bmatrix} \in \mathbb{R}^6, \quad r_p \in \mathbb{R}^3 \text{ its position, and } v_p \in \mathbb{R}^3 \text{ its velocity,}$$

having a close encounter with a secondary uncontrolled satellite

$$x_s = \begin{bmatrix} r_s \\ v_s \end{bmatrix} \in \mathbb{R}^6, \quad r_s \in \mathbb{R}^3, \quad v_s \in \mathbb{R}^3.$$

The dynamics of the two objects are given by:

$$\frac{dx_p}{dt} = F(x_p) + \epsilon G(x_p)u, \tag{1}$$

$$\frac{dx_s}{dt} = F(x_s), \tag{2}$$

where $F(\cdot) : \mathbb{R}^6 \to \mathbb{R}^6$ in Eq. (1) denotes the drift, representing uncontrolled motion of the objects. $F(\cdot)$ can include any dynamics, though in this work we limit the drift to two-body acceleration only. $\epsilon \in \mathbb{R}$ denotes the maximum acceleration due to thrust, $G(x_p) : \mathbb{R}^6 \to \mathbb{R}^{6\times 3}$ the influence of control on the dynamics, and $u \in \mathcal{U} \subset \mathbb{R}^3$ the normalised control, representing thrust acceleration. In this work, we express u in RSW-components, meaning that $G(\cdot)$ involves a frame transformation from RSW to ECI (in which the state vector is expressed).

Our primary focus is on the *time of maximum danger*, denoted by t_{MD}, at which the chosen danger metric is evaluated. In this work, we consider only *short encounters*, meaning that the danger metric is only assessed at t_{MD}, rather than in an entire time interval. It is therefore sufficient to design a collision avoidance manoeuvre which changes the danger metric to safety at t_{MD} only.

To evaluate the safety of a close approach, three commonly used metrics are often considered. The first is the Euclidean distance, defined as

$$d_E = \|r_p - r_s\|, \tag{3}$$

where r_p and r_s are the positions of the primary and secondary satellites, respectively, and $\|\cdot\|$ denotes the Euclidean norm.

The second commonly used metric is the Mahalanobis distance, which accounts for the shape and orientation of the positional uncertainty ellipsoids and is defined as

$$d_M = \sqrt{(r_p - r_s)^\top \Sigma^{-1} (r_p - r_s)}. \tag{4}$$

The Mahalanobis distance provides a normalised measure of separation that incorporates both the relative distance and the positional uncertainties. These metrics offer complementary perspectives on collision risk and are used depending on the context and precision required.

The third metric is the probability-of-collision formula, which estimates the likelihood of collision based on the relative position and positional uncertainty. For two satellites with Gaussian-distributed positional uncertainties, the probability of collision can be expressed as

$$d_{PoC} = \frac{R^2}{2\pi\sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2} d_M^2\right), \tag{5}$$

making use of the definition of the Mahalanobis distance d_M, where $\Sigma = \Sigma_p + \Sigma_s$ is the combined covariance matrix, with Σ_p and Σ_s representing the covariance matrices of the primary and secondary satellites, respectively. R is the combined hard-body radius representing the physical size of the objects.

From now on, $d_{(\cdot)}$ represents the danger of distance metric of a conjunction, without specifying the exact metric choice.

Optimisation Problem

Regardless of the danger metric used, the goal of a Collision Avoidance Maneuver (CAM) is to reduce this metric to a safe threshold at t_{MD}, thereby ensuring that the threshold is not exceeded for any t during short encounters. The objective is to maximise the time available before initiating the CAM, which is equivalent to minimising $-t_s$, where t_s is the (negative) start time of the manoeuvre:

$$
\begin{aligned}
&\min_{||u||\leq 1} -t_s \quad \text{subject to:} \\
&\quad\begin{cases} \dfrac{dx_p}{dt} = F(x) + \varepsilon G(x)\, u, \quad \forall t \in [t_0, t_{MD}], \\[2ex] u \in \mathcal{U} \in \mathbb{R}^3, \quad \forall t \in [t_0, t_{MD}], \end{cases} \\
&x_p(t_s) = x_{p,t_s}, \\
&d_{(\cdot)}(t_f) = \sigma_{(\cdot)}, \\
&t_f = t_{MD}(t_f),
\end{aligned}
\tag{6}
$$

where t_s and t_f denote the start and end times of the problem (and the CAM thrust arc), respectively. Here, t_{MD} is nominally assumed to be zero, meaning that t_s is negative, and minimising $-t_s$ corresponds to maximising the thrust initiation time. The initial state $x_p(t_s)$ is determined by propagating the natural drift of the primary satellite backward in time from the current state. The parameter σ represents the safety threshold, which depends on the specific danger metric employed. Operationally, values like $\sigma_E = 1$ km as a minimum Euclidean distance or $\sigma_{PoC} = 10^{-4}$ as a maximum PoC are often used.[6] Finally, the constraint $t_f = t_{MD}(t_f)$ ensures that the final time of the optimisation problem coincides with the *actual* time of maximum danger, which may vary due to the control input. The thrust arc is then ensured to cover the entire time interval $[t_s, t_{MD}]$, with t_{MD} variable under control.

To account for the variation of t_{MD} with the control input, we leverage an approximation based on the Euclidean distance metric. At the point of closest approach, the relative position and velocity vectors are orthogonal:

$$
(r_p(t_{MD}) - r_s(t_{MD})) \cdot (v_p(t_{MD}) - v_s(t_{MD})) = 0, \tag{7}
$$

where the relative position vector and the relative velocity vector are orthogonal at the moment of geometric closest approach. While this orthogonality condition holds strictly for the Euclidean distance, we use it as a first-order approximation for t_{MD} when using the PoC or Mahalanobis distance, recognising that these metrics also incorporate uncertainty information, as the geometric closest approach often aligns closely with the time of maximum danger. For small uncertainties, these metrics are dominated by the relative distance, validating the use of orthogonality as a first-order approximation.

DIFFERENTIAL ALGEBRA APPROACH

In order to use differential algebra effectively in the context of the optimisation problem shown in Eq. (6), it is convenient to discretise the time grid between t^{alert} (the time at which a dangerous conjunction in the future is noticed) and $\bar{t}_{MD}$ (the nominal time of maximum danger), where $t^{alert} < t_s < t_{MD}$. Obtained is a discretised time grid $t_0, t_1, ..., t_N$ of N intervals, where $t_0 = t^{alert}$ and $t_N = \bar{t}_{MD}$.

We consider $r_p(t_{N-1})$ and $v_p(t_{N-1})$, the state of the primary satellite at time t_{N-1}, as:

$$r_p(t_{N-1}) = \bar{r}_p(t_{N-1}) + \delta r_p(t_{N-1}), \tag{8}$$

$$v_p(t_{N-1}) = \bar{v}_p(t_{N-1}) + \delta v_p(t_{N-1}), \tag{9}$$

that is to say, they are vectors described by the sum of their nominal state ($\bar{r}_p(t_{N-1})$ and $\bar{v}_p(t_{N-1})$) and possible perturbations ($\delta r_p(t_{N-1})$ and $\delta v_p(t_{N-1})$) caused by control u. The state of the secondary object is unchanged:

$$r_s(t_{N-1}) = \bar{r}_s(t_{N-1}), \quad v_s(t_{N-1}) = \bar{v}_s(t_{N-1}),$$

as the control u does not act on the secondary object.

We consider a possible thrust duration from $t_0 = t^{alert}$ to $t_N = \bar{t}_{MD}$, assuming that the optimal time of starting thrust t_s is within this range. To facilitate the DA implementation, we introduce a time transformation:

$$\tau = \frac{t}{T} \tag{10}$$

, where $T = \bar{t}_{MD} - t_0$, leading to $\frac{d}{dt} = \frac{d}{d\tau}\frac{1}{T}$. Note that T can vary under the influence of control, as the time of maximum danger t_{MD} is strongly dependent on the orbit of the primary satellite. We can therefore write

$$T(t_{N-1}) = \bar{t}_{MD} + \delta t = t_{MD} \tag{11}$$

for some perturbation δt. For simplicity, we initially assume a constant thrust profile $u([t_{N-1}, \bar{t}_{MD} + \delta t]) = 0 + \delta u([t_{N-1}, \bar{t}_{MD} + \delta t])$ in RSW components in within each time interval: a zero nominal thrust profile and a thrust perturbation δu.

The state at $t_{MD} + \delta t$ is computed by propagating the perturbed state at t_{N-1} using a Taylor series expansion (obtained via DA) that incorporates the effects of the control perturbations (δu) and the time perturbation (δt): Eq. (12) maps the perturbed state of the primary satellite at time t_{N-1} to the nominal time of maximum danger $\bar{t}_{MD}$, where $\mathcal{T}(\cdot)$ represents the Taylor series expansion obtained from the DA-based propagation. This approach allows us to express the system's behaviour as Taylor polynomials in terms of perturbations δr_p, δv_p, δu and δt.

$$\begin{cases} r_p(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r_p(t_{N-1}), \delta v_p(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{MD} + \delta t]), \delta t) \\ v_p(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r_p(t_{N-1}), \delta v_p(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{MD} + \delta t]), \delta t) \\ r_s(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta t) \\ v_s(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta t) \\ r_{\text{rel}}(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r_p(t_{N-1}), \delta v_p(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{MD} + \delta t]), \delta t) \\ v_{\text{rel}}(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r_p(t_{N-1}), \delta v_p(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{MD} + \delta t]), \delta t) \\ \Delta\tau(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r_p(t_{N-1}), \delta v_p(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{MD} + \delta t]), \delta t) \end{cases} \tag{12}$$

Here, $r_{\text{rel}} = r_p - r_s$ and $v_{\text{rel}} = v_p - v_s$ are defined as the relative position and velocity respectively, in the case of Eq. (12) at time $\bar{t}_{MD} + \delta t$, and $\Delta\tau$ the change in thrust duration as a result of the aforementioned perturbations. Note that δr_p and δv_p cover perturbations at t_{N-1}, δu covers a perturbation in the interval $[t_{N-1}, \bar{t}_{MD} + \delta t]$ and δt covers a perturbation at the end of the time interval $[t_{N-1}, \bar{t}_{MD}]$. The state of the secondary satellite is unaffected by control, and the Taylor polynomials describing the state in Eq. (12) are therefore only dependent on δt.

To ensure the control-induced perturbations accurately mitigate collision risk, δt must satisfy the closest approach condition, where the relative position and velocity vectors are orthogonal. The time perturbation δt is determined by enforcing the orthogonality condition (Eq. (13)) at perturbed time $\bar{t}_{MD} + \delta t$, linking perturbations at t_{N-1} to the geometric closest approach:

$$\delta t = \delta t_{CA} = \mathcal{T}(\delta r(t_{N-1}), \delta v(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{CA} + \delta t])) \tag{13}$$

The orthogonality condition $r_{\text{rel}} \cdot v_{\text{rel}} = 0$ at t_{MD} ensures that the derived δt corresponds to the time when the relative distance between the objects is minimal. This guarantees the perturbations are accurately aligned with the dynamics of the conjunction. Resolving δt ensures that all evaluations at $\bar{t}_{MD} + \delta t$ track the moment of maximum danger.

Next, the danger metric at $\bar{t}_{MD} + \delta t$ (tracking t_{MD}) can be approximated by a Taylor polynomial like in Eq. (12). Note that all danger metrics described in Eq. (3), Eq. (5) and Eq. (4) are primarily dependent on r_{rel}. We therefore also have

$$d_{(\cdot)}(\bar{t}_{MD} + \delta t) = \mathcal{T}(\delta r_p(t_{N-1}), \delta v_p(t_{N-1}), \delta u([t_{N-1}, \bar{t}_{MD} + \delta t]), \delta t) \tag{14}$$

To determine the optimal control perturbation δu, we expand the danger metric $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$ around the nominal value:

$$d_{(\cdot)}(\bar{t}_{MD} + \delta t) = d_{(\cdot)}(\bar{t}_{MD}) + \frac{\partial d_{(\cdot)}}{\partial \delta u}\delta u([t_{N-1}, \bar{t}_{MD} + \delta t]) + \mathcal{O}(||\delta u||^2) \tag{15}$$

In low-thrust propulsion scenarios, the limited thrust magnitude ($\epsilon << 1$) necessitates precise targeting of the danger metric's rate of change to achieve optimal collision avoidance. To maximise the rate of change of the danger metric, the control perturbation δu, is chosen to be aligned with the gradient:

$$\delta u([t_{N-1}, \bar{t}_{MD} + \delta t]) = \epsilon \frac{\frac{\partial d_{(\cdot)}}{\partial \delta u}}{||\frac{\partial d_{(\cdot)}}{\partial \delta u}||} \tag{16}$$

Eq. (16) can be substituted in Eq. (14), together with $\delta r_p(t_{N-1}) = \delta v_p(t_{N-1}) = 0$ to investigate the effect of $\delta u([t_{N-1}, \bar{t}_{MD} + \delta t])$ on $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$. There are now two cases possible:

- If $d_{(\cdot)}(\bar{t}_{MD} + \delta t)$ exceeds the safety threshold $\sigma_{(\cdot)}$, we refine the solution within the interval t_{N-1}, t_N to obtain a more precise value of t_s

- $d_{(\cdot)}(\bar{t}_{MD}+\delta t)$ has not yet reached the safety threshold $\sigma_{(\cdot)}$. In that case, we consider the time interval $[t_{N-2}, t_{N-1}]$ and obtain the following system:

$$\begin{cases} r_p(t_{N-1}) = \mathcal{T}(\delta r_p(t_{N-2}), \delta v_p(t_{N-2}), \delta u([t_{N-2}, t_{N-1}]), \delta t) \\ v_p(t_{N-1}) = \mathcal{T}(\delta r_p(t_{N-2}), \delta v_p(t_{N-2}), \delta u([t_{N-2}, t_{N-1}]), \delta t) \\ r_s(t_{N-1}) = \mathcal{T}(\delta t) \\ v_s(t_{N-1}) = \mathcal{T}(\delta t) \end{cases} \tag{17}$$

  Root-finding can be used to obtain δt again, ensuring a tracking of the closest approach, and $r_p(t_{N-1})$, $v_p(t_{N-1})$, $r_s(t_{N-1})$ and $v_s(t_{N-1})$ can be substituted in Eq. (14), together with Eq. (16), obtaining $\delta u([t_{N-2}, t_{N-1}])$. The same process can now be repeated for previous time steps.

A backward sweep is therefore performed over the time intervals $[t_{N-1}, t_N + \delta t]$, $[t_{N-2}, t_{N-1}]$, ..., $[t_0, t_1]$ until the safety threshold $\sigma_{(\cdot)}$ is reached, depending on the exact danger metric, and t_s can be found accordingly.

The idea of the methodology is sketched out in Figure 1, with a nominal trajectory in blue along discretised time steps t_j, $j \in [0, N]$ Starting at the time interval $[t_{N-1}, t_N]$, where $t_N = \bar{t}_{MD}$

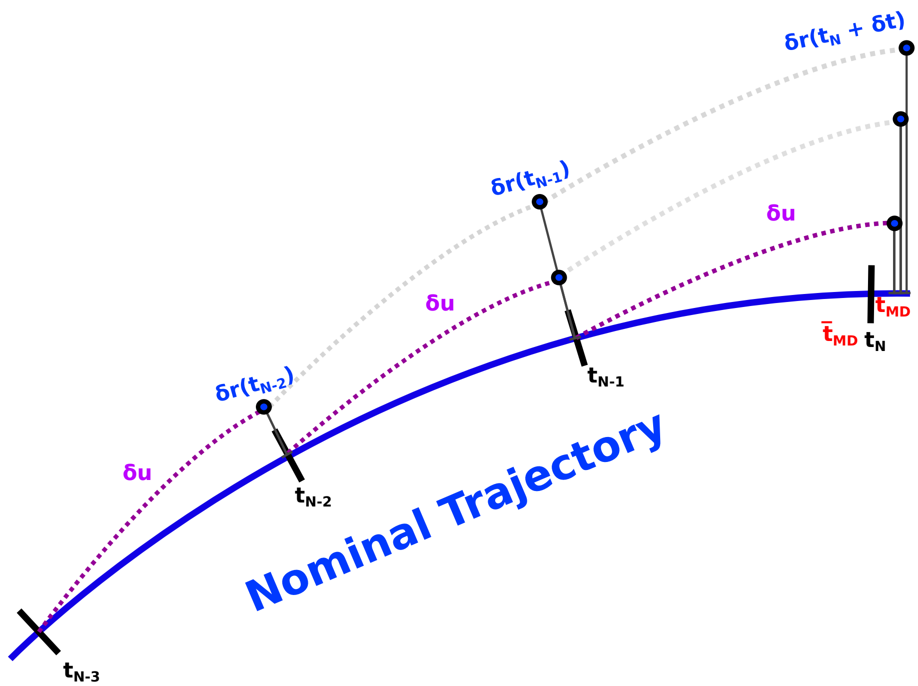

Figure 1. Illustration of the backward sweep procedure using Differential Algebra. The nominal trajectory is shown in blue, with discrete time steps. Perturbations in position (δr), velocity (δv), and control (δu) are used to iteratively refine the manoeuvre start time (t_s).

the nominal time of maximum danger, the position deviation $\delta r(t_N + \delta t)$ is tracked, depending on $\delta r_p(t_{N-1})$, $\delta v_p(t_{N-1})$ (not explicitly shown), δu in the time interval $[t_{N-1}, t_N + \delta t]$, and δt itself. δt can be found through polynomial inversion with Eq. (13) and δu through rate-of-change maximisation of $d_{(\cdot)}$ in Eq. (16). In this first time interval, $\delta r_p(t_{N-1})$ and $\delta v_p(t_{N-1})$ are zero, enabling computation of $\delta r_p(t_N + \delta t)$; the lower one of the right three black dots in Figure 1. Going back one time interval to $[t_{N-2}, t_{N-1}]$ and performing the same method, $\delta r_p(t_{N-1})$ and $\delta v_p(t_{N-1})$ are obtained. Their non-zero values are substituted in the Taylor polynomial for $\delta r_p(t_N + \delta t)$, which now shifts to the second of the right three black dots in Figure 1 under the influence of control in the time interval $[t_{N-2}, t_{N-1}]$. The same can be done for the time interval $[t_{N-3}, t_{N-2}]$, shown by the upper of the three black dots in Figure 1.

APPLICATIONS

While the proposed methodology is valid for a wide range of satellite conjunctions, the geometry of the conjunction can play a significant role, not only in the control profile $u(t)$, but also in the methodology accuracy. As mentioned, the control profile requires discretisation in the DA methodology described in this work and is assumed to have constant components in the time intervals. Therefore, when the optimal control profile changes rapidly, the discretised control might be less accurate. Additionally, the combined covariance of the primary and secondary objects changes over time for non-circular orbits, increasing near the periapsis and decreasing near the apsis.

In this section, the algorithm is applied to a representative case study of a typical ISS-like orbit, having close to zero eccentricity in Low-Earth Orbit. Simulation parameters are shown in Table 1.

Table 1. Simulation parameters

Parameter	Case Study (ISS-like)
Semi-major axis, a (km)	6780
Eccentricity, e	10^{-3}
Inclination, i (deg)	51.6
RAAN, Ω (deg)	0
Argument of Perigee, ω (deg)	0
True Anomaly, ν at $\bar{t}_{MD}$ (deg)	30
Thrust Magnitude, ϵ (m/s^2)	$1.0 \cdot 10^{-4}$
Gravitational Parameter, μ (m^3/s^2)	$3.986 \cdot 10^{14}$
Number of Nodes, N	50
Alert Time, t^{alert} (s)	-1800
Nominal miss distance, d_E (m)	200
Safe miss distance, σ_E (m)	1000
Nominal standard deviation, σ_x (m)	100
Nominal covariance, ρ_{xy} (m^2)	0.4
Nominal standard deviation, σ_y (m)	75
Safe PoC, σ_{PoC}	10^{-5}
Safe Mahalanobis distance, σ_M (-)	3

In the Results section, the control profiles $u(t)$ are investigated for the case study and the different danger metrics d_E, d_{PoC} and d_M, and the change in danger metrics over time is also shown.

RESULTS

The discretised control profile $u(t)$ of the first case study with an ISS-like orbit is shown in Figure 2, for three different distance metrics:

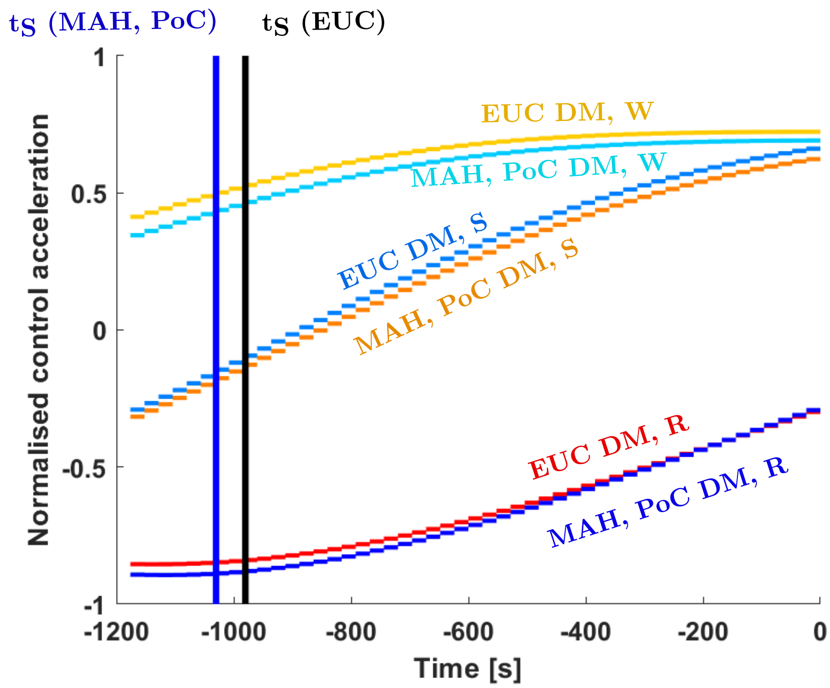

Figure 2. The control profile (radial, along-track and cross-track components) over time for the Euclidean, Mahalanobis and PoC distance metrics, and the latest time to initiate thrust

The discrete nature of the control in Figure 2 is clear, as the control components are constant within the discrete time intervals. The Mahalanobis and PoC-based control profiles are nearly identical, as both depend heavily on the covariance matrix Σ. They differ slightly from the control profile based on the Euclidean distance danger metric, but not significantly. Vertical lines in Figure 2 indicate the maximum t_s values for each distance metric. These values are not the same for the three danger metrics, as d_M and d_{PoC} depend to a large extent on the combined covariance Σ, while d_E does not.

Additionally, the change of the Euclidean distance danger metric throughout the backward sweep is shown in Figure 3.

The thrust duration is shown in Figure 3, rather than the time t itself, coinciding with $t_{MD} - t_s$. t_{MD} varies with time in principle, but the change from its nominal value is less than a second and not visible in Figure 2 and Figure 3. Since the control is unconstrained in direction, the Euclidean distance metric is continuously increasing with thrust until the safe threshold is reached.

CONCLUSION AND DISCUSSION

The introduced DA methodology provides several advantages for low-thrust CAM optimisation. First, the control profile minimising $-t_s$ is quickly determined using a first-order approximation. Given the small thrust magnitudes involved, this first-order solution is often sufficiently accurate. If needed, it can be refined to second-order accuracy by extending the Taylor expansion of the danger metric $d_{(\cdot)}$. Furthermore, the methodology is flexible, allowing the use of various danger metrics beyond the three mentioned in this work. Additionally, operational constraints, such as restricted thrust intervals or directional limitations, can be seamlessly integrated into the DA framework.

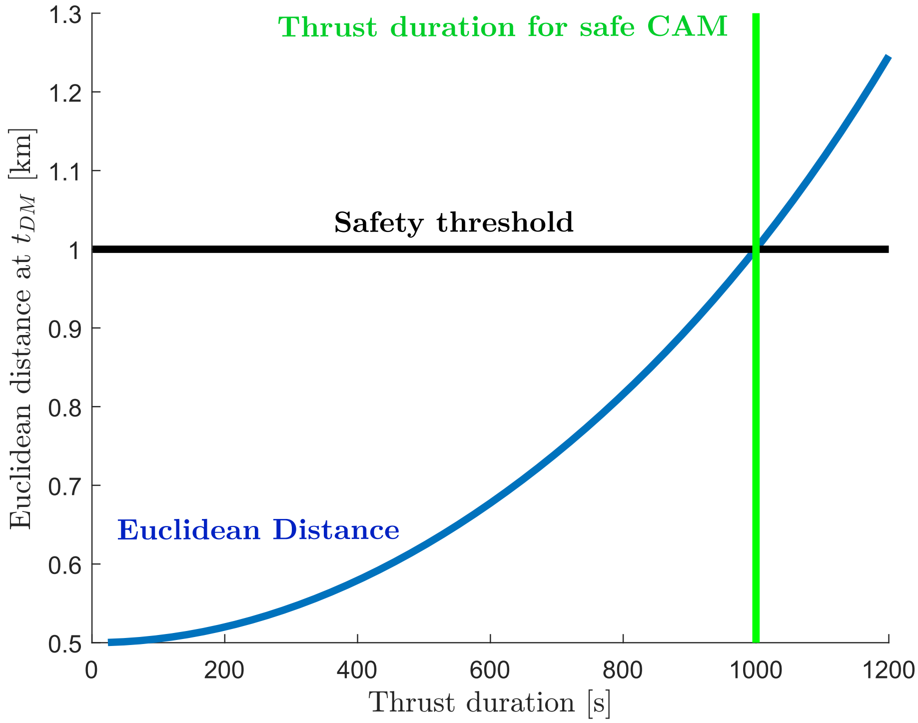

Figure 3. The Euclidean distance metric for various thrust durations

One of the methodology's key strengths is its explicit tracking of t_{MD}, the time of maximum danger. This is achieved through polynomial inversion of the δt term, not only in the time interval $[t_{N-1}, \bar{t}_{MD} + \delta t]$ but also in earlier intervals. During the backward sweep, t_{MD} is accurately updated through the Taylor polynomial representing $r_{\text{rel}}(\bar{t}_{MD} + \delta t)$, ensuring that the danger metric $d_{(\cdot)}$ is evaluated at the correct time. However, the control profile $u_{t_{N-1} \to \bar{t}_{MD} + \delta t}$ is not recalculated in earlier intervals. In other words, once δt is resolved for $[t_{N-1}, \bar{t}_{MD} + \delta t]$, the thrust continues uninterrupted until the recalculated t_{MD}. Earlier control profiles $u_{t_{i-1} \to t_i}$ $(i = 1, 2, \ldots, N-1)$ adjust t_{MD} as δt evolves, but they do not retroactively modify the control in subsequent intervals.

Despite these advantages, the methodology has some limitations. The control profile u must be discretised over time, with constant components in each time interval. Choosing an appropriate discretisation number N can be challenging, and modelling errors may be introduced if the true optimal control profile u^* (found e.g. using optimal control theory) varies rapidly within an interval. Additionally, the methodology heavily depends on two key assumptions: (1) that the time of maximum danger coincides with the geometric closest approach, and (2) that thrust magnitudes remain small relative to the natural dynamics. As a result, this approach is unsuitable for impulsive CAMs or long encounters with small $||v_{\text{rel}}||$ values between the primary and secondary.

ACKNOWLEDGMENTS

The authors would like to express their gratitude to the developers of the DACE library for C++ for providing software that enabled efficient differential algebra computations utilised in this work. The authors also extend their appreciation to Thales Alenia Space for their sponsorship of a portion of the PhD program under which this research was conducted. Furthermore, the authors acknowledge the support of Université Côte d'Azur for funding a research stay in Auckland, New Zealand, which greatly contributed to the progress of this work.

REFERENCES

[1] L. Dell'Elce, F. d. Veld, and J.-B. Pomet, "Characterization of the minimum warning time for low-thrust collision avoidance maneuvers," *Astrodynamics specialist conference*, 2024.

[2] Z. Pavanello and L. Pirovano, "Recursive Polynomial Method for Fast Collision Avoidance Maneuver Design," *arXiv preprint*, 2024.

[3] A. Mahfouz, G. Gaias, and H. Voos, "Fuel-optimal formation reconfiguration by means of unidirectional low-thrust propulsion system," *Journal of Guidance, Control, and Dynamics*, 2025.

[4] C. Paduraru and A. Solomon, "Collision Avoidance and Return Manoeuvre Optimisation for Low-Thrust Satellites using Reinforcement Learning," *ResearchGate*, 2024.

[5] Z. Pavanello, L. Pirovano, R. Armellin, and A. D. Vittori, "Collision Avoidance Maneuvers Optimization in the Presence of Multiple Encounters," *arXiv preprint*, 2024.

[6] K. G. Symonds, T. Flohrer, N. Mardle, D. Fornarelli, X. Marc, and T. Ormston, "Operational reality of collision avoidance manoeuvres," *SpaceOps 2014 Conference*, 2014, p. 1746.